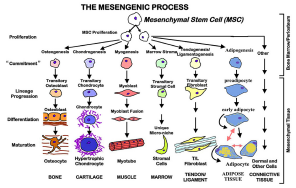
# Computational Biology

## Lecture 15: *Cell Differentiation*

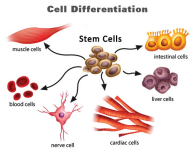



*Random Boolean networks, stem cells, epigenetics*



# Outline

1. Cell Differentiation
2. The Life of an Organism
3. Epigenetic






# Cell Specialisation

- Unlike unicellular organisms most multi-cellular organisms exhibit different types of cells
- By differentiating, cells can be much better adapted to a particular task
- In humans there are around 200 different types of cells
- Each cell contains the same DNA so cell differentiation involves reading a different set of instructions

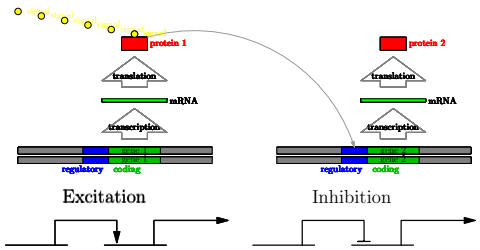


# Cell Functions

- Cell function is determined by the expression of different genes
- The expression of genes are determined by the transcription factors (proteins that interact with the promoter sites either exciting or inhibiting the transcription of a gene)
- Cells can respond to their environment by the interaction between signalling molecules and proteins
- Thus, each gene type will respond differently to its environment



# Regulation by Signal



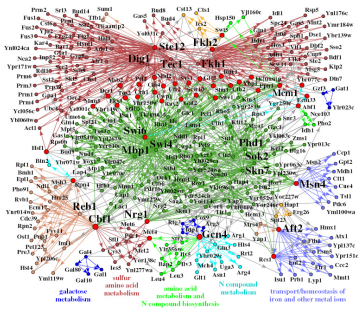



# Gene Regulation Networks

- Proteins control each other's expression forming complex gene regulation networks
- The pattern of interaction determines the type of cell
- Once the type is determine cells overwhelming stay in that type



# Large Gene Regulation Network



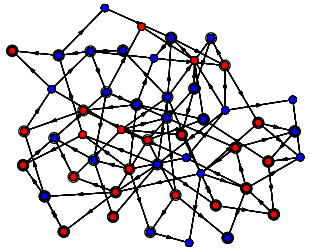


# Random Boolean Networks

| $k$ | $f_k(F,F)$ | $f_k(F,T)$ | $f_k(T,F)$ | $f_k(T,T)$ |
|---|---|---|---|---|
| 0 | F | F | F | F |
| 1 | T | F | F | F |
| 2 | F | T | F | F |
| 3 | T | T | F | F |
| 4 | F | F | T | F |
| 5 | T | F | T | F |
| 6 | F | T | T | F |
| 7 | T | T | T | F |
| 8 | F | F | F | T |
| 9 | T | F | F | T |
| 10 | F | T | F | T |
| 11 | T | T | F | T |
| 12 | F | F | T | T |
| 13 | T | F | T | T |
| 14 | F | T | T | T |
| 15 | T | T | T | T |

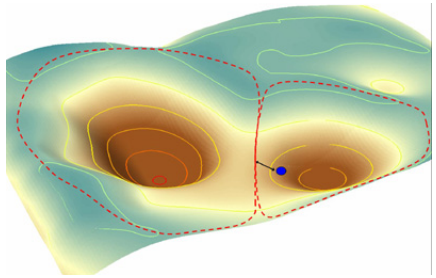
- To understand gene dynamics researchers (lead by Stuart Kauffman) considered the dynamics of random Boolean networks
- Each node is connected to $k_{in}$ other nodes uniformly at random
- The nodes perform a randomly selected Boolean functions

## Dynamics of RBN





## Attractors



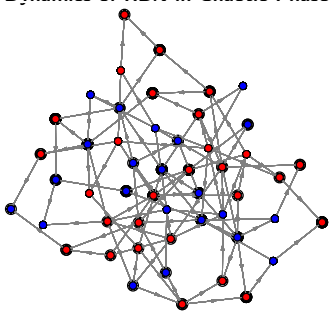


## Phase Transition

- The behaviour of random Boolean networks shows two phases depending on the number of connections and the randomness of the functions
  - For sparsely connected networks (in-degree $\leq 2$) there are a small number of attractors which are reached after a small number of iterations
  - Otherwise there are an exponential number of attractors which take exponential time to reach—the "chaotic" phase
- It has been argued that biology lives on the "edge of chaos" where there are only a few attractors, but the network is adaptable



## Dynamics of RBN in Chaotic Phase





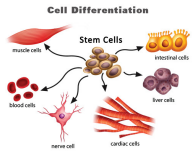
## Real Biology

- Of course, biological systems are more complex
- It is very difficult to turn one type of cell into another
- When cells divide they faithfully remember what type they are
- There are influences beyond the current protein interactions that affect a cell's type (epigenetics)



## Outline

1. Cell Differentiation
2. **The Life of an Organism**
3. Epigenetic






## In the Beginning

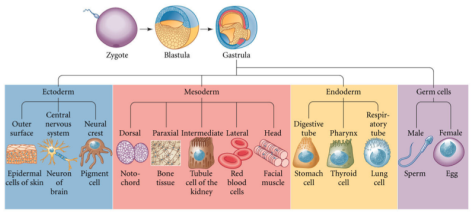
- The life cycle of multi-cellular organisms are complex
- They begin life as a single embryo cell
- This cell will divide to form all the specialised cells of the organism (and the placenta)
- The embryo cell can make any other cell and is known as totipotent
- The embryo grows through cell division and quickly differentiates into cells of the organism and cells of the placenta



## Embryonic Stem Cells

- The cells in the organism are no longer able to form a placenta, but can produce any other cell in the body
- These **embryonic stem cells** are said to be **pluripotent**
- The embryonic stem cells can change to more specialised stem cells which are said to be *multipotent*
- These will eventually specialise to the normal cells of the body known as the **soma**

## Specialisation

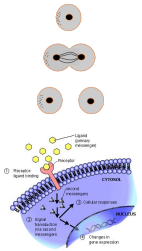



Gilbert 9th Edition Fig. 1.7

Somatic vs. Germline cell lineages



## Causes of Specialisation



- There are a number of routes to specialisation
- During the development of the embryo some transcription factors aggregate, at cell division this leads to differentiation in the cell function
- Other cells are known to specialise depending on signalling molecules that occur at receptors



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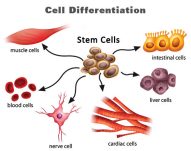


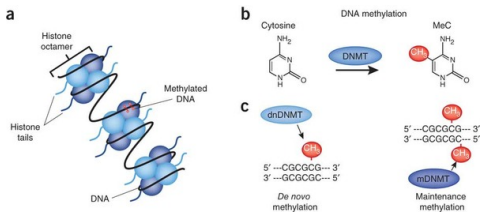



## Cell Division

- Cells in the body constantly divide
- They all carry the same DNA code
- But, the DNA is often modified by methylation and acetylation
- These can be facilitated, e.g. by proteins called *DNA methyltransferase* (DNMT)

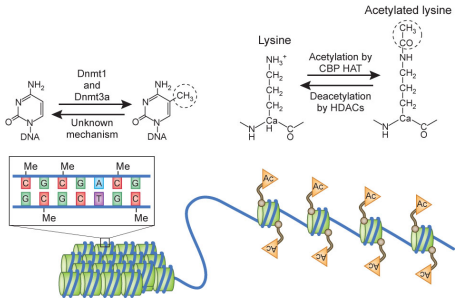


## DNA Methylation






## Histone Acetylation






## Epigenetics

- Acetylation suppresses the expression of genes
- DNA methylation can suppress or activate the expression of genes
- Stem cells have more methylated sites than somatic (normal) cells, showing that this is an important mechanism for ensuring cell specialisation is maintained during cell division
- The methylation and acetylation is modified by proteins
- Epigenetics can be seen as another component in a complex Boolean network



## Conclusion

- Cell differentiation is a complex biological process that is essential to multi-cellular organism
- Simplified models of Boolean networks suggests that a small number of attractors might naturally occur in an interacting network
- Whether this "explains" multi-cellular organisms or their origin is a matter of debate
- Specialisation happens through a hierarchical series of steps
- Epigenetics seem to be an important component in determining which proteins are expressed